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Potential is “intrinsic energy of inertia”

label

$$\nabla \cdot \epsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \epsilon_0 \frac{\partial}{\partial t} \mathcal{E} \mathcal{B} = \nabla \times U_m \quad \mathcal{E} = -\nabla U_e - \frac{\partial}{\partial t} U_m \quad (1)$$

⇒ Wave equation for potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (2)$$

⇒ Potential by “ignition J” (cause, inhomogeneity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (3)$$

elec. Potential structure:

$$J = -\frac{1}{\epsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\epsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} It \quad (4)$$

magn. Potential structure:

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \epsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const.}}{=} \quad (5)$$

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (6)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (7)$$

Electric induced energy (exogenous excitation) $E_{kin} = \frac{1}{2}mv^2 \hat{=} (pc)^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \epsilon_0 \mathcal{E}^2 \quad (8)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (9)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (10)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (11)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (12)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \quad (13)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (14)$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (15)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (16)$$

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogeneity, cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \quad C = \text{Capacitance} \quad Q = \text{charge (ion)} \quad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{radiation intensity}$$

$$D_S = \int (\cdot \mathcal{S}) \, dr = \int \operatorname{div}(\mathcal{E} \times \mathcal{H}) \, dr = \text{dissipation energy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda\frac{1}{T} = \sqrt{\frac{k}{m}} \hat{=} kc \hat{=} \frac{1}{\sqrt{LC}} \hat{=} \sqrt{\frac{g}{r}} = \text{frequency}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor (dielectricum)}$$

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Low Frequency:

$$\omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \quad (17)$$

$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

High Frequency:

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (18)$$

$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity in Vacuum

Continuity in Matter

Refraction index emerges

Near Field

Far Field

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with Dielectricum, Refraction emerges, Reflection, Absorption

Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\alpha) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{aligned} \quad (19)$$

¹Wirkungsquantum

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (20)$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \quad (21)$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \quad e = \text{electron charge} \quad Q = \text{ion charge}$$

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

$$\Rightarrow \text{Ionisation Energy of chemical element } E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$$

$$\Rightarrow \text{Energy released (gained) } E_{gain} = E_{pot} - E_i = mc^2 - h\nu$$